

KABPRO Subscription Data Model

Audience: Engineering Status: Catalog + assignment implemented - Entitlement enforcement planned

1. Goals

- Single catalog of sellable plans with country pricing
- Per-organization subscription with trial lifecycle
- Separate marketing benefits from machine entitlements
- Enforce access in Admin UI + API (future phases)
- Support seat types, add-ons, and per-org overrides without forking catalog rows

2. Implemented models

2.1 SubscriptionPlan

File: server/src/models/SubscriptionPlan.js Rule: one row per (code, country)

Field	Type	Notes
name	String	Display name
code	String	Uppercase, indexed
category	Enum	Free, Starter, Professional, Growth, Enterprise, Custom
description	String	Short summary
benefits	[{ title, detail }]	Marketing bullets
features	[String]	Legacy - currently synced from benefit titles (not entitlement keys yet)
country	String	ISO 3166-1 alpha-2
countryName	String	Display
currency	String	INR, AED, USD, ?
priceMonthly	Number	
priceYearly	Number	
pricingLabel	String	e.g. Free, Custom
trialDays	Number	Default 14
limits.vehicles	Number	Soft quota (not enforced yet)
limits.drivers	Number	Soft quota
limits.users	Number	Soft quota (seats)
featured	Boolean	
isActive	Boolean	
sortOrder	Number	

Indexes: unique { code, country }; { country, category, sortOrder }

2.2 OrganizationSubscription

File: server/src/models/OrganizationSubscription.js

Field	Type	Notes
organization	ObjectId -> Organization	
plan	ObjectId -> SubscriptionPlan	

status	Enum	Trial, Active, PastDue, Cancelled, Expired
billingCycle	Enum	Monthly, Yearly, Custom
trialStartsAt / trialEndsAt	Date	
startsAt / endsAt	Date	
notes	String	
assignedBy	ObjectId -> User	Superadmin actor

Missing today (planned): payment IDs, invoices, entitlement snapshot, seat quantities purchased, add-on refs.

2.3 Supporting platform models

Model	Purpose
Organization	Platform customer
Project	Linked project / agency context
PlatformLead	Sales pipeline
PlatformEvent	Audit / funnel trail

2.4 Legacy note

Project.plan free-text enum (Trial / Starter / ?) is not the source of truth. Prefer OrganizationSubscription.

3. Planned model extensions (P0+)

3.1 Entitlement feature entries on plan

Replace legacy string features with structured keys (keep benefits separate):

```
entitlements: [{
  key: String,          // e.g. 'departments.monthly_billing'
  enabled: Boolean,
  value: mongoose.Schema.Types.Mixed // optional number/enum
}]
```

3.2 Seat packs

```
seatTypes: [{
  role: { type: String, enum: ['admin', 'manager', 'operator', 'viewer'] },
  included: Number,
  priceExtra: Number // per extra seat / month in plan currency
}]
```

3.3 Trial subset (optional)

```
trialEntitlements: [String] // if empty -> same as paid entitlements
```

3.4 Org entitlement override (new collection)

```
OrgEntitlementOverride {
  organization: ObjectId,
  enable: [String],      // force-on keys
  disable: [String],     // force-off keys
  limits: { ... },       // override caps
  notes: String,
  updatedBy: ObjectId
}
```

3.5 Add-ons (new collections)

```
PlanAddOn {
  code, name, description,
  country, currency,
  priceMonthly, priceYearly,
  entitlements: [...],
  quotaDeltas: { users, vehicles, ... }
}
```

```
OrgAddOn {
  organization, addOn, quantity,
  status, startsAt, endsAt
}
```

3.6 Entitlement snapshot on subscription

When assigning / converting, copy resolved entitlements onto the subscription document so catalog edits do not silently change live customers until renewal (policy choice - document and stick to it).

4. Entitlement resolution algorithm

```
function resolveEntitlements(org):
  sub = active OrganizationSubscription for org
  if !sub or sub.status in [Cancelled]: return readOnlyMinimal
  if sub.status == Expired: return readOnlyMinimal

  base = plan.entitlements (+ trialEntitlements if Trial)
  add  = union OrgAddOn entitlements
  over = OrgEntitlementOverride

  features = (base ? add ? over.enable) - over.disable
  limits   = merge(plan.limits, add.quotaDeltas, over.limits)

  return { features, limits, sub, plan, benefits: plan.benefits }
```

Expired / no sub: allow login + billing page + read-only existing data; block creates.

5. API design

5.1 Superadmin (existing + extend)

Method	Path	Notes
GET	/api/superadmin/plans	Filter country, category, active
POST	/api/superadmin/plans	Upsert (body.id for update)
PUT	/api/superadmin/plans/:id	Update
DELETE	/api/superadmin/plans/:id	Block if active/trial subs
POST	/api/superadmin/plans/seed-defaults	IN/AE/US templates
GET	/api/superadmin/subscriptions	List
POST	/api/superadmin/subscriptions	Assign (+ startTrial)
PATCH	/api/superadmin/subscriptions/:id	Status / notes

Extend upsert body to accept entitlements[], seatTypes[], trialEntitlements[].

5.2 Fleet Admin (new)

Method	Path	Notes
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GET	/api/billing/me	Resolved plan, benefits, features map, limits, usage, trialEndsAt
GET	/api/billing/catalog	Optional public/active plans for upgrade UI (country from org)
POST	/api/billing/upgrade-request	Creates lead / notifies Superadmin

5.3 Middleware (new)

```
requireEntitlement('departments.monthly_billing')
requireQuota('vehicles') // before Vehicle.create
```

Return 403 with machine code:

```
{
  "success": false,
  "error": "Plan limit reached",
  "code": "QUOTA_VEHICLES",
  "limit": 25,
  "usage": 25,
  "upgradeRequired": true
}
```

6. Admin client contract

GET /api/billing/me response shape (target):

```
{
  "success": true,
  "subscription": {
    "status": "Trial",
    "billingCycle": "Monthly",
    "trialEndsAt": "2026-09-30T00:00:00.000Z",
    "daysLeft": 14
  },
  "plan": {
    "name": "Starter",
    "code": "STARTER",
    "country": "IN",
    "currency": "INR",
    "priceMonthly": 2999,
    "benefits": [{ "title": "...", "detail": "..." }]
  },
  "features": {
    "module.dashboard": true,
    "departments.monthly_billing": false,
    "drivers.payroll": false
  },
  "limits": { "vehicles": 25, "drivers": 50, "users": 5 },
  "usage": { "vehicles": 12, "drivers": 20, "users": 3 }
}
```

Admin wraps app in EntitlementProvider:

- can('departments.monthly_billing')
- Hide/disable sidebar items
- Show upgrade CTA on locked routes
- Trial banner when status === 'Trial'

7. Trial automation (P3)

Job	Action
Daily cron	Find Trial with trialEndsAt < now -> set Expired; emit event; notify
Reminder job	D-7 / D-3 / D-1 -> in-app notification to primary admin
Grace (optional)	PastDue for N days after failed renewal before Cancelled

8. Seed defaults (current)

Markets: IN (INR), AE (AED), US (USD) Templates: Starter, Professional, Growth, Enterprise with sample monthly/yearly prices and benefit bullets.

Command path: Superadmin UI Seed markets -> POST /plans/seed-defaults.

After P0, seed should also write entitlement keys per tier (see matrix in 02).

9. Migration notes

- Existing plans keep working; add entitlements with defaults derived from category
- Keep writing legacy features string array from benefit titles until Admin no longer needs it for display
- Backfill OrganizationSubscription for orgs missing a row (onboard already creates Trial)
- Do not delete Project.plan until UI references are removed

10. Testing checklist

- [] Create plan IN + US same code, different prices
- [] Duplicate (code, country) returns 409
- [] Delete blocked while Trial/Active/PastDue exists
- [] Assign trial sets trialEndsAt = now + trialDays
- [] GET /billing/me returns features + usage
- [] Locked module returns 403 with upgradeRequired
- [] Over-quota vehicle create returns QUOTA_VEHICLES
- [] Cron expires trial and Admin shows wall
- [] Org override enables a single Professional feature on Starter

11. Related documents

- 01-ADMIN-FEATURES - module -> entitlement keys
- 02-PLAN-ENTITLEMENTS-BLUEPRINT - commercial matrix & roadmap
- 03-SUPERADMIN-PLATFORM - operator workflows

End of Subscription Data Model